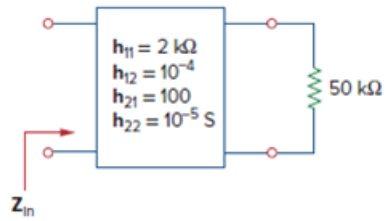


Problem

Find the impedance at the input port of the circuit in Fig. 19.27.

Figure 19.27

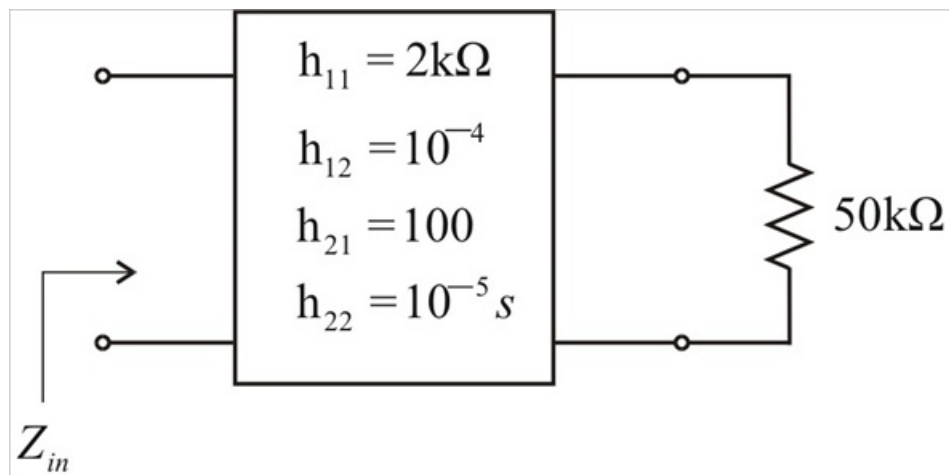


Step-by-step solution

Step 1 of 4

Given two port circuit is

Step 2 of 4



Step 3 of 4

The hybrid parameters based on V_1 and I_2 the dependent variables. Thus,

$$V_1 = h_1 I_1 + h_{12} V_2$$

$$I_2 = h_2 I_1 + h_{22} V_2$$

Substituting the hybrid parameters, we obtain

$$V_1 = 2000 I_1 + 10^{-4} V_2 \text{ ----- (1)}$$

$$I_2 = 100 I_1 + 10^{-5} V_2 \text{ ----- (2)}$$

Since from the circuit the output voltage at output port

$$V_2 = -50000 I_2$$

Substituting in equation (2), we get

$$I_2 = 100 I_1 + 10^{-5} (-50000 I_2)$$

$$I_2 (1 + 0.5) = 100 I_1$$

$$I_2 = \frac{100}{1.5} I_1$$

Step 4 of 4

Substituting values of V_2 and I_2 in equation (1), we obtain

$$V_1 = 2000 I_1 + 10^{-4} (-50000) I_2$$

$$V_1 = 2000 I_1 + (-5) \left(\frac{1000}{15} \right) I_1$$

$$V_1 = I_1 \left[2000 - \frac{1000}{3} \right]$$

$$\frac{V_1}{I_1} = \left(\frac{5}{3} \right) 1000$$

Thus, the impedance at the input port of the circuit

$$\therefore \boxed{Z_{in} = 1.6667 \text{ k}\Omega}$$